Staffing Decision Report

Should you replace **sale sp12**? · Branded Clothing Store, Bangalore

✔ REPLACE sale sp12 (+■6.74 expected gross margin over 6 months)

Executive Summary

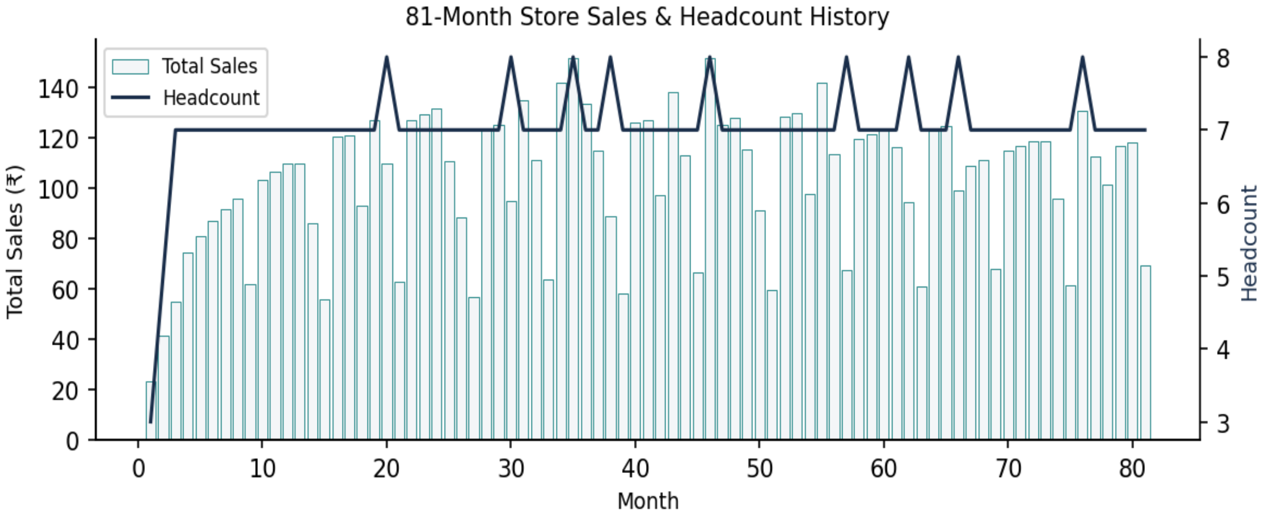
sale sp12 ranks **#4 out of 16** salespeople who have ever worked here (~80th percentile), with a skill effect of **+1.80 ■**/month after adjusting for tenure ramp-up and seasonality. Replacing her with a new hire of typical quality is projected to add **■6.74** in gross margin over the next 6 months versus leaving the seat empty. Replacing is the better call unless the new hire turns out to be in roughly the bottom 20% of historical hiring outcomes.

Key Metrics at a Glance

Metric	Value
Data span	81 months
Total salespeople (ever)	16
sale sp12 — skill rank	#4 / 16 (80th pct.)
sale sp12 — skill effect	+1.80 ■/month
Expected margin gain from replacing	■+6.74 (6 mo)
Break-even new-hire quality	~20th pct. of historical hires
Seasonal cycle detected	12 months
Model R^2	0.924
Capacity ceiling evidence	Yes — 8th staff cannibalises ~4.8 ■ sales/month

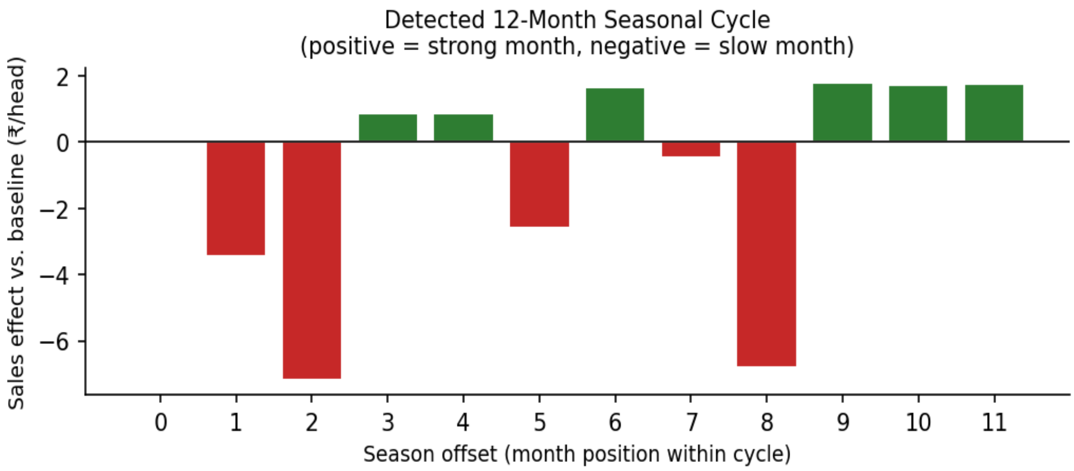
1. Store History: Sales & Headcount

The store has been running for 81 months. Normal staffing level is 7 people; brief spikes to 8 occur during 1-month overlap hiring. Total monthly sales broadly follow an upward trend with clear seasonal variation.



2. Seasonality

A statistically meaningful **12-month repeating cycle** was detected in per-head sales. Two distinct slow periods occur roughly every 6 months (likely monsoon-related and post-festive-season lulls, common in Bangalore malls). All performance comparisons are seasonality-adjusted.

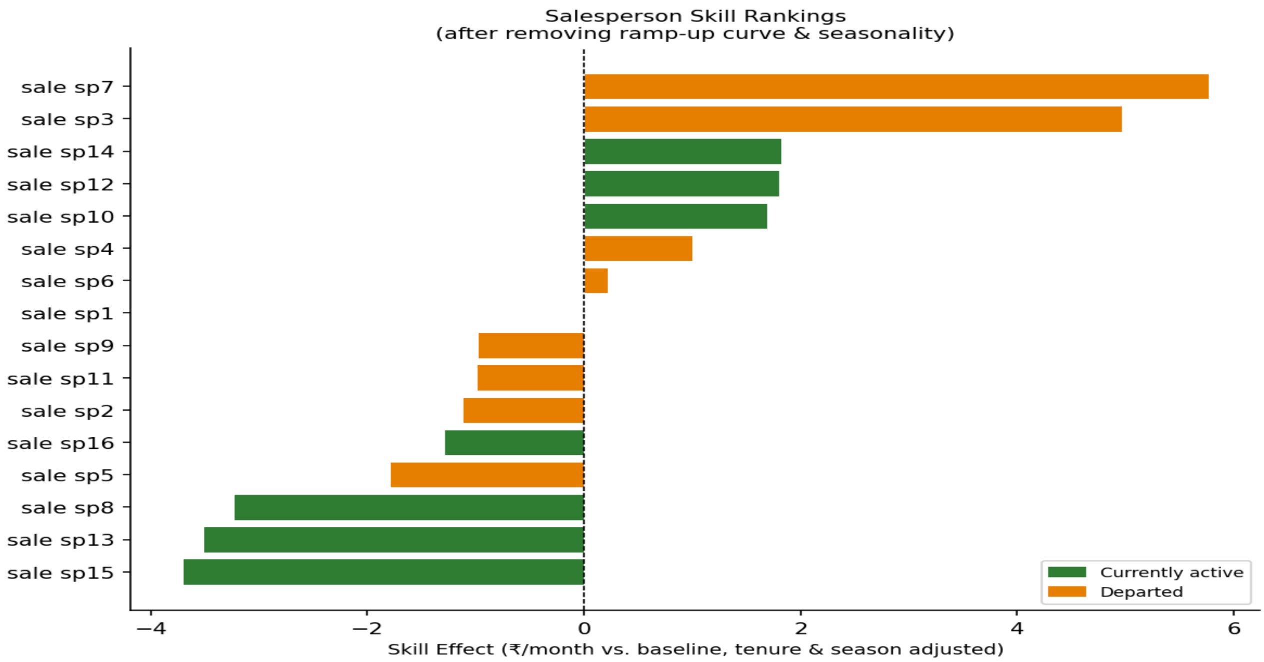


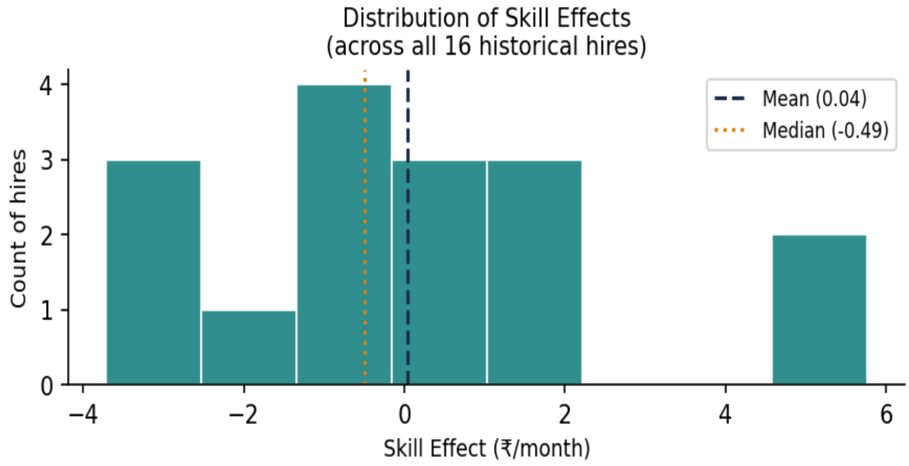
3. Salesperson Skill Rankings (Tenure & Season Adjusted)

Each salesperson's 'skill effect' is their estimated monthly sales contribution versus a baseline hire, *after* removing how long they'd been on the job (ramp-up curve) and which calendar season it was. This is the fairest basis for comparing people who started at different times.

Rank	Salesperson	Start	End	Tenure (mo)	Skill Effect	Status
1	sale sp7	3	57	55	5.77	Departed
2	sale sp3	1	46	46	4.97	Departed
3	sale sp14	62	81	20	1.82	Active
4	sale sp12	46	81	36	1.8	Active
5	sale sp10	35	81	47	1.69	Active
6	sale sp4	2	35	34	1.0	Departed
7	sale sp6	3	38	36	0.22	Departed
8	sale sp1	1	66	66	0.0	Departed
9	sale sp9	30	76	47	-0.98	Departed
10	sale sp11	38	62	25	-0.99	Departed
11	sale sp2	1	30	30	-1.12	Departed
12	sale sp16	76	81	6	-1.29	Active
13	sale sp5	2	20	19	-1.79	Departed
14	sale sp8	20	81	62	-3.24	Active
15	sale sp13	57	81	25	-3.52	Active
16	sale sp15	66	81	16	-3.71	Active

Highlighted row = sale sp12 (resigning). Green = currently active. Yellow highlight = person under review.





Distribution of skill effects across all 16 hires. A new hire's quality is unknown in advance; the scenarios below use this distribution to bound the uncertainty.

4. Store Capacity Ceiling Test

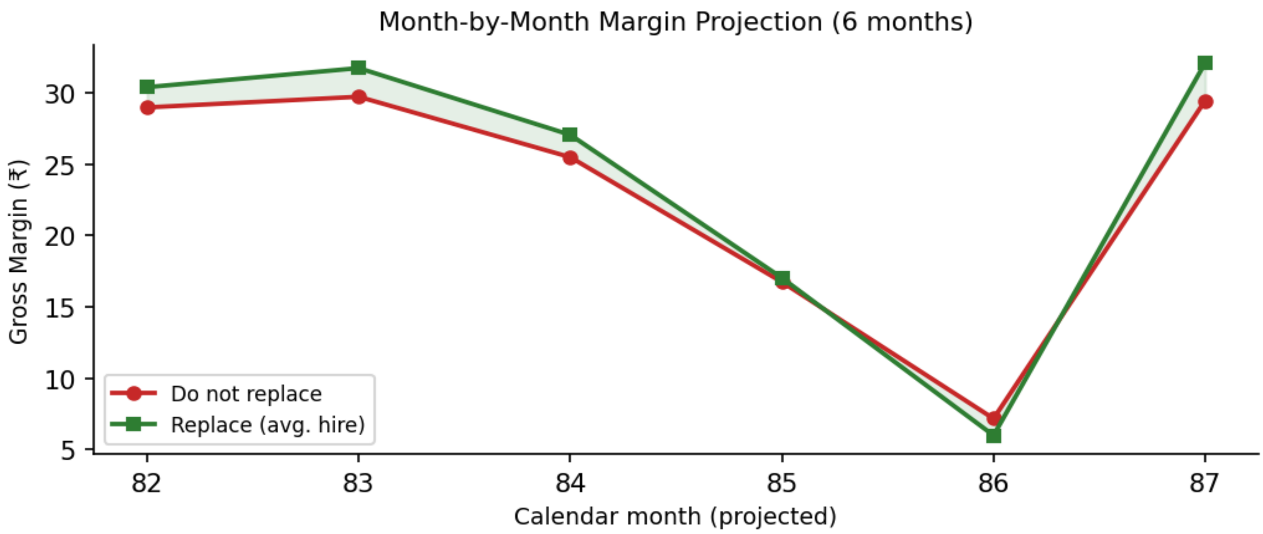
During the 9 months when headcount was 8 (overlap-hire months), total store sales were on average **-7.7 ■** below what the established staffers' individual track records would predict (vs. **-0.08** in normal 7-person months). This 7.7 ■ gap suggests a **customer footfall ceiling**: an 8th salesperson displaces sales from existing colleagues rather than generating net-new business.

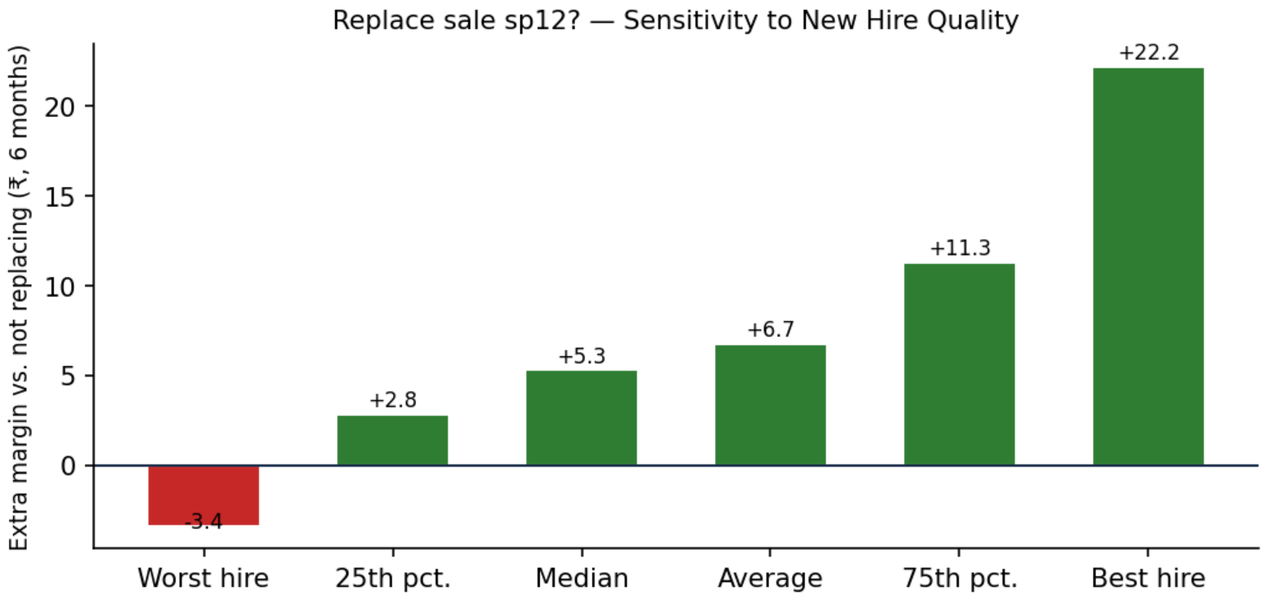
> Recommendation on overlap-hiring: Time the new hire's start to land close to the departing person's last day, not a full month early. Caveat: the sales data cannot capture the training/handover value of overlap months — if you find new hires ramp faster after overlapping with a predecessor, that value may offset the margin cost.

5. Financial Comparison: Replace vs. Do Not Replace

Margin formula: Gross Margin = 50% × Sales – ■3.0 × headcount – 5% × Sales = 45% × Sales – ■3.0 × headcount

Scenario	Skill (■/mo)	6-mo Margin (■)	vs. Not Replacing (■)
Do NOT replace	—	137.46	—
Replace — worst historical hire (bottom)	-3.71	134.08	-3.38
Replace — 10th percentile hire	-3.38	134.96	-2.50
Replace — pessimistic hire (25th pct.)	-1.42	140.27	+2.81
Replace — median hire	-0.49	142.77	+5.31
Replace — average hire (recommended baseline)	+0.04	144.2	+6.74
Replace — optimistic hire (75th pct.)	+1.72	148.74	+11.28
Replace — 90th percentile hire	+3.40	153.27	+15.81
Replace — best historical hire	+5.77	159.66	+22.20





Bars above zero = replacing is better. Only the worst-historical-hire outcome (far left) makes not replacing the better call.

6. Rules of Thumb: Revised Recommendations

■ Replace whenever a salesperson resigns — with one exception

Replacing is better than leaving a seat empty for any new hire of median or better quality (which is the most likely outcome). Only hold off if you have specific reason to believe the pool of available candidates is unusually weak.

■ Do not reflexively overlap for a full month

Your data shows an 8th salesperson during overlap months generates ~■4.8 less total store sales than the 7 established staffers' combined track records would predict. Time the new hire's start to coincide with — or just after — the departing person's last day, unless there is a specific handover task that genuinely requires side-by-side time.

■ Re-run this analysis every time someone leaves

The right answer depends on who is leaving (their skill rank), the upcoming season, and who remains on the team. The script included in this package will recompute everything from your latest CSV in seconds.

■ Track the skill ranking as a management tool

The tenure- and season-adjusted skill table ranks every current and past hire fairly. Use it to identify underperformers early (before they reach a natural resignation point) and to benchmark new hires at the 6-month and 12-month marks.

■ Be cautious about running 8 staff permanently

Current evidence suggests your store has a customer footfall ceiling at roughly 7 staff. Adding an 8th person on a sustained basis is unlikely to grow total sales meaningfully; it will mostly add ■3/month in salary cost.

Assumptions & Caveats

- Sales units are treated as a consistent currency across all 81 months.
- Each salesperson's employment is one continuous block of months (verified in data).
- A new hire's future skill is estimated from the historical distribution of all 16 past hires' adjusted skill effects, since the actual new hire is unknown.
- The detected seasonal pattern continues into the next 6 months unchanged.
- The margin formula (50% gross margin, 5% commission, ■3/month salary) is as given and applied uniformly across all staff.
- The model explains ~92% of month-to-month variation in individual sales (R^2=0.92), giving high confidence in the skill rankings and projections.
- Training/handover value of overlap months is not captured in sales numbers and may partly offset the observed capacity-ceiling penalty.